Assignment 009 (B01 to B18) (B19 to B52) (B31 to B50)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>DeFi Protocol Lab Guide</title>

    <!-- Tailwind CSS -->
    <script src="https://cdn.tailwindcss.com"></script>

    <!-- Chart.js -->
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

    <!-- Google Fonts -->
    <link
href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700&display=swap"
rel="stylesheet">

    <script>
        tailwind.config = {
            theme: {
                extend: {
                    fontFamily: {
                        sans: ['Inter', 'sans-serif'],
                    },
                    colors: {
                        gray: {
                            50: '#F9FAFB',
                            100: '#F3F4F6',
                            800: '#1F2937',
```

```
                900: '#111827',
            },
            accent: {
                light: '#E0F2FE', // sky-100
                main: '#0284C7',  // sky-600
                dark: '#0369A1',  // sky-700
            },
            success: '#10B981', // emerald-500
        }
    }
  }
}
</script>

<style>
  /* Required Chart Container Styling */
  .chart-container {
    position: relative;
    width: 100%;
    max-width: 800px;
    margin-left: auto;
    margin-right: auto;
    height: 300px;
    max-height: 400px;
  }
  @media (min-width: 768px) {
    .chart-container {
      height: 350px;
    }
  }
```

```
    /* Custom Scrollbar */
    ::-webkit-scrollbar {
        width: 8px;
    }
    ::-webkit-scrollbar-track {
        background: #F3F4F6;
    }
    ::-webkit-scrollbar-thumb {
        background: #D1D5DB;
        border-radius: 4px;
    }
    ::-webkit-scrollbar-thumb:hover {
        background: #9CA3AF;
    }

    /* Utility for hiding elements */
    .js-hidden {
        display: none !important;
    }

    /* Interactive list item styling */
    .checklist-item {
        transition: all 0.2s ease-in-out;
        cursor: pointer;
    }
    .checklist-item.checked {
        opacity: 0.6;
        background-color: #f0fdf4; /* Light green bg */
        border-color: #86efac;
    }
</style>
```

<!-- Chosen Palette: Warm Neutrals with Sky Blue Accent -->

<!-- Application Structure Plan: A tabbed instructional dashboard. This structure is chosen because the source report is a sequential laboratory assignment. Presenting it as a linear "course" with distinct sections (Overview, Tools, Architecture, and a Mock Simulation) breaks down technical instructions into digestible steps, preventing cognitive overload. The "Mock Simulation" section adds a crucial interactive layer, allowing the user to experience the end-goal of the code they are supposed to write. -->

<!-- Visualization & Content Choices:

1. Overview: Text -> Goal: Inform -> Clean typography layout.

2. Setup & Tools: List -> Goal: Organize -> Interactive Checklist (HTML/JS) -> Allows users to track physical progress.

3. Architecture: Flow -> Goal: Organize/Relationships -> CSS Grid Block Diagram -> Visualizes the abstract relationship between components (User -> Network -> Contract) without using external images/SVG.

4. Simulation: Interactive UI -> Goal: Experience/Change -> Custom HTML Form + Chart.js Line Chart -> Simulates the Solidity contract logic (deposit/withdraw math) and visualizes the hypothetical Total Value Locked (TVL) changing in real-time based on user input, providing immediate visual feedback on the concepts being taught.

-->

<!-- CONFIRMATION: NO SVG graphics used. NO Mermaid JS used. -->

</head>

<body class="bg-gray-50 text-gray-800 font-sans antialiased min-h-screen flex flex-col">

<!-- Header -->

<header class="bg-white border-b border-gray-200 shadow-sm sticky top-0 z-10">

⚒ <!-- Hammer/Pick icon using Unicode -->

<h1 class="text-xl font-bold text-gray-900 tracking-tight">DeFi Lab: Mock Lending Contract</h1>

Testnet Simulation Environment

```
    </div>
  </header>

  <!-- Main Layout -->
  <div class="flex-grow max-w-7xl mx-auto w-full flex flex-col md:flex-row gap-6 p-4 sm:p-6 lg:p-8">

    <!-- Navigation Sidebar -->
    <nav class="md:w-64 flex-shrink-0 bg-white rounded-xl shadow-sm border border-gray-200 p-4 h-fit">
      <h2 class="text-xs font-semibold text-gray-400 uppercase tracking-wider mb-4 px-2">Lab Modules</h2>
      <ul class="space-y-1" id="nav-menu">
        <li>
          <button data-target="section-overview" class="nav-btn w-full flex items-center gap-3 text-left px-3 py-2.5 rounded-lg text-sm font-medium transition-colors bg-accent-light text-accent-main">
            <span>&#x1F4DD;</span> 1. Objective & Overview
          </button>
        </li>
        <li>
          <button data-target="section-tools" class="nav-btn w-full flex items-center gap-3 text-left px-3 py-2.5 rounded-lg text-sm font-medium text-gray-600 hover:bg-gray-50 transition-colors">
            <span>&#x1F6E0;</span> 2. Tools & Setup
          </button>
        </li>
        <li>
          <button data-target="section-architecture" class="nav-btn w-full flex items-center gap-3 text-left px-3 py-2.5 rounded-lg text-sm font-medium text-gray-600 hover:bg-gray-50 transition-colors">
            <span>&#x1F4CA;</span> 3. Protocol Architecture
          </button>
        </li>
        <li>
```

```
            <button data-target="section-simulation" class="nav-btn w-full flex items-center gap-3 text-left px-3 py-2.5 rounded-lg text-sm font-medium text-gray-600 hover:bg-gray-50 transition-colors">
                <span>&#x1F3AE;</span> 4. Interactive Simulation
            </button>
        </li>
    </ul>
</nav>

<!-- Content Area -->
<main class="flex-grow bg-white rounded-xl shadow-sm border border-gray-200 p-6 sm:p-8 min-h-[600px]">

    <!-- SECTION 1: OVERVIEW -->
    <section id="section-overview" class="content-section fade-in">
        <div class="mb-8 border-b border-gray-100 pb-4">
            <h2 class="text-3xl font-bold text-gray-900 mb-2">Objective</h2>
            <p class="text-gray-600 text-lg leading-relaxed">
                To interact with a decentralized finance (DeFi) protocol on a blockchain testnet by writing and executing a smart contract that deposits and withdraws tokens, demonstrating basic DeFi operations without real funds.
            </p>
        </div>

        <div class="bg-gray-50 rounded-xl p-6 border border-gray-100">
            <h3 class="text-xl font-semibold text-gray-800 mb-4 flex items-center gap-2">
                <span class="text-accent-main">&#x1F4A1;</span> Practical Description
            </h3>
            <p class="text-gray-700 mb-4 leading-relaxed">
                In this practical, we simulate interaction with a DeFi protocol by creating a simple lending contract where:
            </p>
            <ul class="space-y-3 mb-6">
```

```
                <li class="flex items-start gap-3">
                  <span class="text-success mt-1">&#x2714;</span>
                  <span class="text-gray-700"><strong>Users can deposit ETH:</strong> Simulating adding liquidity to a pool.</span>
                </li>
                <li class="flex items-start gap-3">
                  <span class="text-success mt-1">&#x2714;</span>
                  <span class="text-gray-700"><strong>Users can withdraw ETH:</strong> Simulating removing liquidity.</span>
                </li>
                <li class="flex items-start gap-3">
                  <span class="text-success mt-1">&#x2714;</span>
                  <span class="text-gray-700"><strong>Contract tracks balances internally:</strong> Ensuring accurate ledger accounting.</span>
                </li>
              </ul>
              <div class="bg-accent-light text-accent-dark p-4 rounded-lg text-sm border border-accent-main/20">
                <strong>Real-World Context:</strong> This represents the foundational mechanics of how major DeFi lending protocols like <em>Aave</em> or <em>Compound</em> operate on underlying testnets before mainnet deployment.
              </div>
            </div>
          </section>

          <!-- SECTION 2: TOOLS & SETUP -->
          <section id="section-tools" class="content-section js-hidden">
            <div class="mb-6">
              <h2 class="text-2xl font-bold text-gray-900 mb-2">Prerequisites & Environment Setup</h2>
              <p class="text-gray-600">Ensure all local environment dependencies are installed before beginning smart contract development. Click items to mark them as ready.</p>
            </div>
```

```
<div class="grid grid-cols-1 md:grid-cols-2 gap-8">
  <!-- Column 1: Local Setup -->
  <div>
    <h3 class="text-lg font-semibold text-gray-800 mb-3 border-b pb-2">Local Development</h3>
    <div class="space-y-3" id="checklist-local">
      <div class="checklist-item border border-gray-200 rounded-lg p-3 flex items-center gap-3 bg-white" onclick="toggleChecklist(this)">
        <div class="w-6 h-6 rounded border-2 border-gray-300 flex items-center justify-center text-transparent font-bold check-icon">&#x2713;</div>
        <div>
          <p class="font-medium text-gray-800">Node.js & npm</p>
          <p class="text-xs text-gray-500">JavaScript runtime & package manager</p>
        </div>
      </div>
      <div class="checklist-item border border-gray-200 rounded-lg p-3 flex items-center gap-3 bg-white" onclick="toggleChecklist(this)">
        <div class="w-6 h-6 rounded border-2 border-gray-300 flex items-center justify-center text-transparent font-bold check-icon">&#x2713;</div>
        <div>
          <p class="font-medium text-gray-800">Visual Studio Code</p>
          <p class="text-xs text-gray-500">Integrated Development Environment</p>
        </div>
      </div>
      <div class="checklist-item border border-gray-200 rounded-lg p-3 flex items-center gap-3 bg-white" onclick="toggleChecklist(this)">
        <div class="w-6 h-6 rounded border-2 border-gray-300 flex items-center justify-center text-transparent font-bold check-icon">&#x2713;</div>
        <div>
          <p class="font-medium text-gray-800">VS Code Extensions</p>
          <p class="text-xs text-gray-500">Install 'Solidity' and 'Hardhat' extensions</p>
        </div>
      </div>
```

```
                <div class="checklist-item border border-gray-200 rounded-lg p-3 flex items-center gap-3 bg-white" onclick="toggleChecklist(this)">
                  <div class="w-6 h-6 rounded border-2 border-gray-300 flex items-center justify-center text-transparent font-bold check-icon">&#x2713;</div>
                  <div>
                    <p class="font-medium text-gray-800">Hardhat Initialization</p>
                    <p class="text-xs text-gray-500">Create a JS/TS project via CLI</p>
                  </div>
                </div>
              </div>
            </div>

            <!-- Column 2: Network Setup -->
            <div>
              <h3 class="text-lg font-semibold text-gray-800 mb-3 border-b pb-2">Blockchain Network</h3>
              <div class="space-y-3" id="checklist-network">
                <div class="checklist-item border border-gray-200 rounded-lg p-3 flex items-center gap-3 bg-white" onclick="toggleChecklist(this)">
                  <div class="w-6 h-6 rounded border-2 border-gray-300 flex items-center justify-center text-transparent font-bold check-icon">&#x2713;</div>
                  <div>
                    <p class="font-medium text-gray-800">MetaMask Wallet</p>
                    <p class="text-xs text-gray-500">Browser extension configured for testnets</p>
                  </div>
                </div>
                <div class="checklist-item border border-gray-200 rounded-lg p-3 flex items-center gap-3 bg-white" onclick="toggleChecklist(this)">
                  <div class="w-6 h-6 rounded border-2 border-gray-300 flex items-center justify-center text-transparent font-bold check-icon">&#x2713;</div>
                  <div>
                    <p class="font-medium text-gray-800">Select Testnet</p>
                    <p class="text-xs text-gray-500">Sepolia (Recommended) / Goerli / Mumbai</p>
```

```
                </div>
              </div>
              <div class="checklist-item border border-gray-200 rounded-lg p-3 flex items-center gap-3 bg-white" onclick="toggleChecklist(this)">
                <div class="w-6 h-6 rounded border-2 border-gray-300 flex items-center justify-center text-transparent font-bold check-icon">&#x2713;</div>
                <div>
                  <p class="font-medium text-gray-800">Acquire Test Tokens</p>
                  <p class="text-xs text-gray-500">Use a faucet to get mock ETH for gas</p>
                </div>
              </div>
            </div>

            <div class="mt-6 p-4 bg-gray-50 rounded-lg border border-gray-200">
              <h4 class="font-medium text-gray-800 text-sm mb-1">Required Deliverables:</h4>
              <ul class="text-sm text-gray-600 list-disc list-inside space-y-1">
                <li>Smart Contract (Solidity)</li>
                <li>Interaction Script (JavaScript)</li>
                <li>Inline logic comments</li>
              </ul>
            </div>
          </div>
        </div>
      </section>

      <!-- SECTION 3: ARCHITECTURE -->
      <section id="section-architecture" class="content-section js-hidden">
        <div class="mb-8">
          <h2 class="text-2xl font-bold text-gray-900 mb-2">Protocol Architecture</h2>
          <p class="text-gray-600">A high-level view of how the tools and code interact to form the decentralized application (dApp) logic.</p>
        </div>
```

```
<!-- Pure HTML/CSS Block Diagram -->
<div class="flex flex-col md:flex-row items-center justify-center gap-4 py-8 overflow-x-auto">

  <!-- Client Side -->
  <div class="flex flex-col items-center">
    <div class="w-48 h-32 bg-white border-2 border-accent-main rounded-xl shadow-sm flex flex-col items-center justify-center p-4 text-center z-10 relative">
      <span class="text-3xl mb-2">&#x1F4BB;</span> <!-- Laptop -->
      <span class="font-bold text-gray-800">Client / Developer</span>
      <span class="text-xs text-gray-500 mt-1">Node.js, Hardhat Script, MetaMask</span>
    </div>
  </div>

  <!-- Connection -->
  <div class="flex flex-col items-center justify-center text-gray-400 font-bold px-2 py-4 md:py-0">
    <span class="hidden md:inline text-2xl">&#x2794;</span> <!-- Right arrow -->
    <span class="md:hidden text-2xl">&#x2B07;</span> <!-- Down arrow -->
    <span class="text-xs text-center mt-1">RPC Call / Transaction</span>
  </div>

  <!-- Network Side -->
  <div class="flex flex-col items-center">
    <div class="w-48 h-32 bg-gray-100 border-2 border-gray-300 border-dashed rounded-xl flex flex-col items-center justify-center p-4 text-center z-10 relative">
      <span class="text-3xl mb-2">&#x1F310;</span> <!-- Globe -->
      <span class="font-bold text-gray-800">Ethereum Testnet</span>
      <span class="text-xs text-gray-500 mt-1">Sepolia Network</span>
    </div>
  </div>
```

```
            <!-- Connection -->
            <div class="flex flex-col items-center justify-center text-gray-400 font-bold px-2 py-4 md:py-0">
              <span class="hidden md:inline text-2xl">&#x2794;</span>
              <span class="md:hidden text-2xl">&#x2B07;</span>
              <span class="text-xs text-center mt-1">Executes</span>
            </div>

            <!-- Contract Side -->
            <div class="flex flex-col items-center">
              <div class="w-48 h-32 bg-accent-light border-2 border-accent-dark rounded-xl shadow-sm flex flex-col items-center justify-center p-4 text-center z-10 relative">
                <span class="text-3xl mb-2">&#x1F4DC;</span> <!-- Scroll -->
                <span class="font-bold text-gray-800">Mock Lending</span>
                <span class="text-xs text-gray-500 mt-1">Solidity Smart Contract (deposit, withdraw)</span>
              </div>
            </div>

          </div>

          <div class="bg-gray-50 p-4 rounded-lg text-sm text-gray-700 mt-4 border border-gray-100">
            <strong>Process Flow:</strong> The developer writes a JavaScript script using Hardhat. This script uses the injected provider (like MetaMask's private key logic) to sign a transaction. This transaction is sent over the internet to the Sepolia Testnet nodes. The network validates the transaction and executes the corresponding function (e.g., <code>deposit()</code>) on the deployed Solidity Smart Contract, altering its internal state (updating the user's balance map).
          </div>
        </section>

        <!-- SECTION 4: INTERACTIVE SIMULATION -->
        <section id="section-simulation" class="content-section js-hidden">
          <div class="mb-6">
```

```
        <h2 class="text-2xl font-bold text-gray-900 mb-2">Lending Protocol Simulation</h2>
        <p class="text-gray-600">Interact with the simulated UI below to see how a user's transaction affects wallet balances and the protocol's internal state (Total Value Locked), fulfilling the assignment's goal.</p>
      </div>

      <div class="grid grid-cols-1 lg:grid-cols-2 gap-8">

        <!-- Left: Interaction Panel -->
        <div class="bg-gray-50 rounded-xl border border-gray-200 p-5 flex flex-col gap-5">

          <!-- Simulated Wallet -->
          <div class="bg-white p-4 rounded-lg border border-gray-200 shadow-sm">
            <div class="flex justify-between items-center mb-2">
              <h3 class="font-semibold text-gray-800 flex items-center gap-2">
                <span class="text-lg">&#x1F45B;</span> Your Wallet (Testnet)
              </h3>
              <span class="text-xs bg-gray-100 text-gray-600 px-2 py-1 rounded">0x71C...9b23</span>
            </div>
            <div class="flex items-end gap-1">
              <span class="text-3xl font-bold text-gray-900" id="wallet-balance">10.00</span>
              <span class="text-gray-500 font-medium mb-1">ETH</span>
            </div>
          </div>

          <!-- Simulated Contract Interface -->
          <div class="bg-white p-4 rounded-lg border border-accent-main shadow-sm relative overflow-hidden">
            <div class="absolute top-0 left-0 w-full h-1 bg-accent-main"></div>
            <h3 class="font-semibold text-gray-800 mb-4 flex items-center gap-2">
              <span class="text-lg">&#x1F3E6;</span> Mock Lending Contract
```

```
                </h3>

                <div class="mb-4">
                  <p class="text-sm text-gray-500 mb-1">Your Deposited Balance:</p>
                  <div class="flex items-end gap-1">
                    <span class="text-2xl font-bold text-accent-main" id="contract-
balance">0.00</span>
                    <span class="text-gray-500 font-medium mb-1">ETH</span>
                  </div>
                </div>

                <!-- Actions -->
                <div class="space-y-3">
                  <div>
                    <label class="block text-xs font-medium text-gray-700 mb-1">Amount
(ETH)</label>
                    <input type="number" id="tx-amount" value="1.0" min="0.1" step="0.1"
class="w-full border border-gray-300 rounded-md shadow-sm py-2 px-3 focus:outline-none
focus:ring-accent-main focus:border-accent-main sm:text-sm">
                  </div>
                  <div class="flex gap-2">
                    <button onclick="handleTransaction('deposit')" class="flex-1 bg-accent-main
hover:bg-accent-dark text-white font-medium py-2 px-4 rounded-md transition-colors focus:outline-
none focus:ring-2 focus:ring-offset-2 focus:ring-accent-main">
                      Deposit
                    </button>
                    <button onclick="handleTransaction('withdraw')" class="flex-1 bg-white
hover:bg-gray-50 text-gray-700 font-medium py-2 px-4 border border-gray-300 rounded-md
transition-colors focus:outline-none focus:ring-2 focus:ring-offset-2 focus:ring-accent-main">
                      Withdraw
                    </button>
                  </div>
                  <p id="tx-message" class="text-xs text-center h-4 text-red-500 mt-2 font-
medium"></p>
```

```
                </div>
            </div>

        </div>

        <!-- Right: Visualization -->
        <div class="flex flex-col">
            <div class="mb-4">
                <h3 class="text-lg font-semibold text-gray-800">Protocol Liquidity (TVL)</h3>
                <p class="text-sm text-gray-500">Tracks total ETH deposited across all simulated users over time.</p>
            </div>

            <!-- Chart Container as per requirements -->
            <div class="chart-container bg-white border border-gray-200 rounded-xl p-4 shadow-sm flex-grow">
                <canvas id="tvlChart"></canvas>
            </div>
        </div>

    </div>
  </section>

 </main>
</div>

<script>
 // --- Navigation Logic ---
 document.querySelectorAll('.nav-btn').forEach(btn => {
  btn.addEventListener('click', function() {
   // Update button styles
   document.querySelectorAll('.nav-btn').forEach(b => {
```

```
            b.classList.remove('bg-accent-light', 'text-accent-main');
            b.classList.add('text-gray-600', 'hover:bg-gray-50');
        });
        this.classList.remove('text-gray-600', 'hover:bg-gray-50');
        this.classList.add('bg-accent-light', 'text-accent-main');

        // Switch sections
        const targetId = this.getAttribute('data-target');
        document.querySelectorAll('.content-section').forEach(sec => {
            sec.classList.add('js-hidden');
        });
        document.getElementById(targetId).classList.remove('js-hidden');
    });
});

// --- Checklist Logic ---
function toggleChecklist(element) {
    element.classList.toggle('checked');
    const icon = element.querySelector('.check-icon');
    if(element.classList.contains('checked')) {
        icon.classList.remove('text-transparent', 'border-gray-300');
        icon.classList.add('text-success', 'border-success', 'bg-green-100');
    } else {
        icon.classList.add('text-transparent', 'border-gray-300');
        icon.classList.remove('text-success', 'border-success', 'bg-green-100');
    }
}

// --- Simulation Logic & Data Storage ---
// State Management
let state = {
```

```
    walletBal: 10.0,
    contractBal: 0.0,
    protocolTVL: 50.0, // Initial fake liquidity from "other users"
    timeLabels: ['Start'],
    tvlData: [50.0]
};

let txCount = 0;

// --- Chart.js Initialization ---
const ctx = document.getElementById('tvlChart').getContext('2d');
const tvlChart = new Chart(ctx, {
    type: 'line',
    data: {
        labels: state.timeLabels,
        datasets: [{
            label: 'Total Value Locked (ETH)',
            data: state.tvlData,
            borderColor: '#0284C7', // accent-main
            backgroundColor: 'rgba(2, 132, 199, 0.1)',
            borderWidth: 2,
            pointBackgroundColor: '#fff',
            pointBorderColor: '#0284C7',
            pointHoverBackgroundColor: '#0284C7',
            pointHoverBorderColor: '#fff',
            fill: true,
            tension: 0.3 // Smooth curves
        }]
    },
    options: {
        responsive: true,
```

```
maintainAspectRatio: false, // Required to respect container height
plugins: {
  legend: {
    display: false
  },
  tooltip: {
    callbacks: {
      label: function(context) {
        let label = context.dataset.label || '';
        if (label) { label += ': '; }
        if (context.parsed.y !== null) {
          label += context.parsed.y.toFixed(2) + ' ETH';
        }
        return label;
      }
    }
  }
},
scales: {
  y: {
    beginAtZero: false,
    suggestedMin: 40,
    grid: {
      color: '#F3F4F6',
      drawBorder: false,
    },
    ticks: { color: '#6B7280' }
  },
  x: {
    grid: { display: false },
    ticks: {
```

```
                    color: '#6B7280',
                    maxRotation: 45,
                    minRotation: 0,
                    // Label wrapping logic implicitly handled by max width, but chart.js usually
truncates.
                }
            }
        },
        interaction: {
            intersect: false,
            mode: 'index',
        },
    }
});

// --- Transaction Handling ---
function updateUI() {
    document.getElementById('wallet-balance').innerText = state.walletBal.toFixed(2);
    document.getElementById('contract-balance').innerText = state.contractBal.toFixed(2);
    document.getElementById('tx-amount').value = ''; // reset input
}

function handleTransaction(type) {
    const amountInput = document.getElementById('tx-amount');
    const msgEl = document.getElementById('tx-message');
    const amount = parseFloat(amountInput.value);

    msgEl.innerText = ''; // clear msg
    msgEl.className = 'text-xs text-center h-4 mt-2 font-medium'; // reset classes

    if (isNaN(amount) || amount <= 0) {
```

```
    msgEl.innerText = 'Enter a valid amount.';
    msgEl.classList.add('text-red-500');
    return;
}

if (type === 'deposit') {
    if (amount > state.walletBal) {
        msgEl.innerText = 'Insufficient Wallet Balance!';
        msgEl.classList.add('text-red-500');
        return;
    }
    // Execute logic
    state.walletBal -= amount;
    state.contractBal += amount;
    state.protocolTVL += amount;
    msgEl.innerText = `Successfully deposited ${amount} ETH.`;
    msgEl.classList.add('text-success');

} else if (type === 'withdraw') {
    if (amount > state.contractBal) {
        msgEl.innerText = 'Insufficient Contract Balance!';
        msgEl.classList.add('text-red-500');
        return;
    }
    // Execute logic
    state.contractBal -= amount;
    state.walletBal += amount;
    state.protocolTVL -= amount;
    msgEl.innerText = `Successfully withdrew ${amount} ETH.`;
    msgEl.classList.add('text-success');
}
```

```
        // Update UI text
        updateUI();

        // Update Chart Data
        txCount++;
        state.timeLabels.push('Tx ' + txCount);
        state.tvlData.push(state.protocolTVL);

        // Keep chart from getting too crowded (keep last 10 points)
        if (state.timeLabels.length > 10) {
            state.timeLabels.shift();
            state.tvlData.shift();
        }

        tvlChart.update();

        // Clear success message after 3 seconds
        setTimeout(() => {
            if(msgEl.innerText.includes('Successfully')) {
                msgEl.innerText = '';
            }
        }, 3000);
    }

    // Init UI
    updateUI();
  </script>
</body>
</html>
```

Output:

Protocol Architecture

A high-level view of how the tools and code interact to form the decentralized application (dApp) logic.

Client / Developer
Node.js, Hardhat Script, MetaMask

→ RPC Call / Transaction

Ethereum Testnet
Sepolia Network

→ Executes

Mock Lending
Solidity Smart Contract (deposit, withdraw)

Process Flow: The developer writes a JavaScript script using Hardhat. This script uses the injected provider (like MetaMask's private key logic) to sign a transaction. This transaction is sent over the internet to the Sepolia Testnet nodes. The network validates the transaction and executes the corresponding function (e.g., `deposit()`) on the deployed Solidity Smart Contract, altering its internal state (updating the user's balance map).

DeFi Lab: Mock Lending Contract

Testnet Simulation Environment

3. Protocol Architecture
4. Interactive Simulation

Local Development

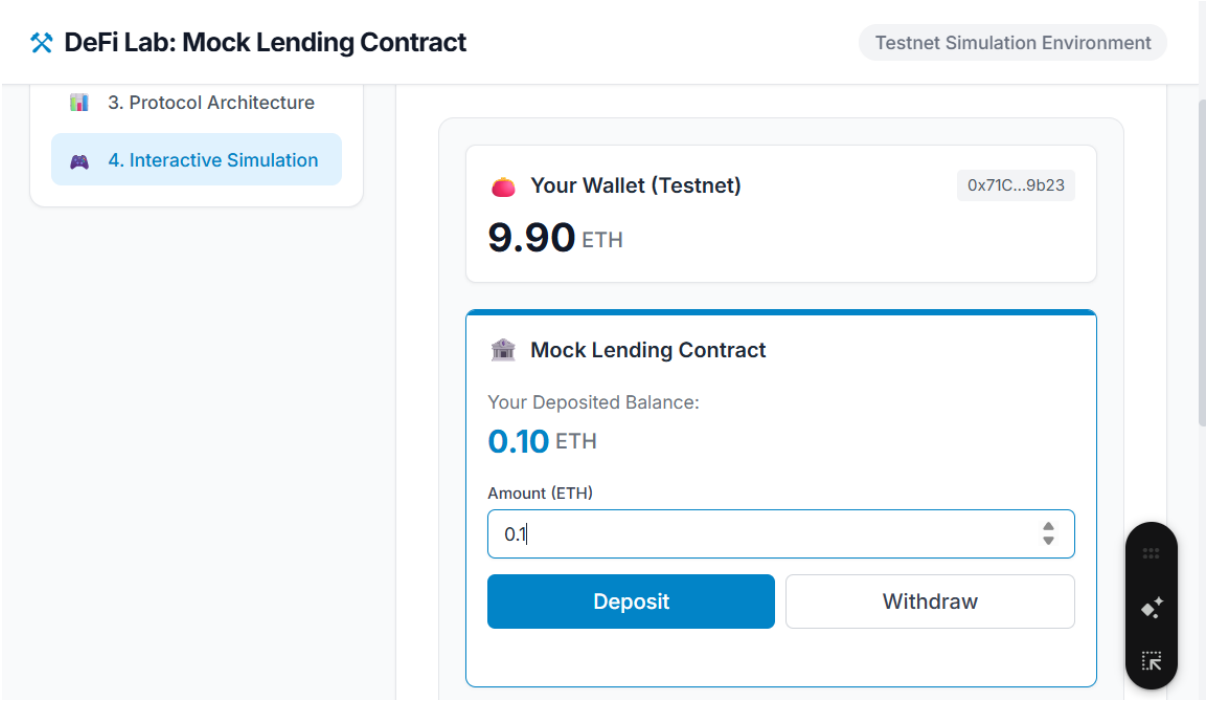
- [x] **Node.js & npm** — JavaScript runtime & package manager
- [x] **Visual Studio Code** — Integrated Development Environment
- [ ] **VS Code Extensions** — Install 'Solidity' and 'Hardhat' extensions
- [ ] **Hardhat Initialization** — Create a JS/TS project via CLI

Blockchain Network

- [x] **MetaMask Wallet** — Browser extension configured for testnets
- [ ] **Select Testnet** — Sepolia (Recommended) / Goerli / Mumbai
- [x] **Acquire Test Tokens** — Use a faucet to get mock ETH for gas

Required Deliverables:

- Smart Contract (Solidity)
- Interaction Script (JavaScript)
- Inline logic comments


DeFi Lab: Mock Lending Contract
Testnet Simulation Environment
3. Protocol Architecture
4. Interactive Simulation
Your Wallet (Testnet)
0x71C...9b23
9.90 ETH
Mock Lending Contract
Your Deposited Balance:
0.10 ETH
Amount (ETH)
0.1
Deposit
Withdraw